

Why Do We Need Sampling?

Imagine a company wants to know:

"What is the average salary of all software engineers in India?"

Can they ask every software engineer?

No.

Why?

- Population is too large.
- It takes too much time.
- It is expensive.
- Some people may not respond.

Instead, they collect information from a **sample**.

Example:

Population = 20 million software engineers

Sample = 1000 software engineers

The sample helps estimate the population characteristics.

Point Estimate

Definition

A **Point Estimate** is a **single numerical value** calculated from a sample that is used to estimate an unknown population parameter.

In simple words:

A **point estimate** is our **best single guess** about the true value of a population parameter based on sample data.

Why is it Called a Point Estimate?

It is called a **point** estimate because it provides **one specific value (a single point)** instead of a range.

For example:

- Population average salary = **Unknown**
- Sample average salary = **₹55,000**

Here, **₹55,000** is the **point estimate** of the population mean.

Example

Suppose a university wants to know the average marks of all 10,000 students.

Instead of asking every student, they randomly select **100 students**.

The average marks of these 100 students is:

75

Since the true population average is unknown, we use **75** as our estimate.

Here,

- Population Mean (μ) = Unknown
- Sample Mean ($\bar{x}$) = **75**

Therefore,

Point Estimate = 75

Problem with Sampling

Different samples give different results.

Example

Population average height = 170 cm

Take Sample 1

- Average = 169

Take Sample 2

- Average = 172

Take Sample 3

- Average = 171

Take Sample 4

- Average = 168

Each sample gives a different answer.

So,

How can we express this uncertainty?

Answer:

Using a **Confidence Interval**.

Confidence Interval (CI)

Definition

A Confidence Interval is a range of values that is likely to contain the true population parameter.

Instead of saying

"The average salary is ₹60,000"

we say

"The average salary is between ₹58,000 and ₹62,000."

This gives a more realistic estimate.

Why Not Give Just One Number?

Suppose

Sample Mean = 75

Is the true population mean exactly 75?

Probably not.

Because the sample contains random variation.

Instead of giving one number,

we provide an interval

72 to 78

This interval has a high chance of containing the actual population mean.

Why do we use a Confidence Interval?

Every sample is different.

Different samples produce different sample means.

Therefore, a single sample mean may not exactly equal the population mean.

A confidence interval accounts for this uncertainty.

What is a Confidence Level?

The **Confidence Level** tells us **how confident we are in the method used to calculate the confidence interval**.

The most common confidence levels are:

Confidence Level	Meaning
90%	Moderate confidence
95%	Most commonly used
99%	Very high confidence

What Does 95% Confidence Mean?

Imagine repeating the same experiment **100 times**.

Each time:

- Take a new random sample.
- Calculate a confidence interval.

Out of those 100 confidence intervals:

- About **95 intervals** will contain the true population mean.
- About **5 intervals** will not.

So, **95% confidence** refers to the **success rate of the method** over many repeated samples.

Visual Representation

None

Population Mean (μ)

Sample 1 -----✓

Sample 2 -----✓

Sample 3 -----✗

Sample 4 -----✓

Sample 5 -----✓

...

100 samples

95 interval contain μ

5 intervals miss μ

Central Limit Theorem (CLT)

Why Do We Need CLT?

Different samples produce different sample means.

Can we predict how those sample means behave?

Yes.

The Central Limit Theorem explains this.

Definition

The Central Limit Theorem states that:

If we repeatedly take random samples of the same size from a population, the distribution of the sample means becomes approximately normal as the sample size becomes sufficiently large, regardless of the original population distribution.

Important Conditions

1. Random sampling.
2. Samples are independent.
3. Sample size is sufficiently large (often $n \geq 30$ is used as a rule of thumb).

Why is CLT Important?

Without CLT

- Normal distribution methods could not be used.

With CLT

- Z-test
- t-test
- Confidence intervals
- Hypothesis testing

all become possible.

Sampling Distribution

A Sampling Distribution is the probability distribution of a statistic (such as the sample mean) obtained from many repeated samples of the same size.

Population



Many Samples



Many Sample Means



Distribution of Sample Means

This distribution is called the **sampling distribution**.

Properties of the Sampling Distribution

The sampling distribution has two important properties:

1. Mean

The mean of all sample means equals the population mean.

$$\mu_{\bar{x}} = \mu$$

Example:

Population average = ₹50,000

Average of all sample means = ₹50,000

2. Spread

The spread of the sampling distribution is measured by the **Standard Error (SE)**.

Standard Error

What is Standard Error?

Standard Error (SE) tells us how much the sample mean changes from one sample to another.

Compare SD and SE

Standard Deviation (SD)	Standard Error (SE)
Variation among individual observations	Variation among sample means
Describes the spread of data	Describes the spread of sample means
Larger SD → Data is more spread out	Smaller SE → Estimates are more precise

Formula

$$SE = \frac{\sigma}{\sqrt{n}}$$

Where:

- **SE** = Standard Error
- **σ** = Population Standard Deviation
- **n** = Sample Size

Example 1

Population SD = **20**

Sample Size = **25**

$$SE = \frac{20}{\sqrt{25}} = \frac{20}{5} = 4$$

This means the sample means typically vary by about **4 units** around the true population mean.

Example 2

Increase the sample size to **100**.

$$SE = \frac{20}{\sqrt{100}} = \frac{20}{10} = 2$$

Now the sample means vary much less.

Key Observation

Sample Size	Standard Error
25	4

100	2
400	1

As the **sample size increases**, the **Standard Error decreases**.

This means:

- Larger sample → Smaller Standard Error
- Smaller Standard Error → More precise estimate
- More precise estimate → Narrower Confidence Interval

Why is CLT Important?

CLT is the foundation of statistical inference because it allows us to:

- Estimate population parameters using samples.
- Construct Confidence Intervals.
- Perform Hypothesis Testing.
- Use the normal distribution for sample means, even when the population isn't normal.

Without CLT, many statistical methods used in data science would not be valid.

Activity

Given the following data:

- Population SD = **18**
- Sample Size = **36**

Calculate the Standard Error.

If the sample size becomes 144, what happens to the Standard Error?